T.A amount allowances and a consistent allocation choice

8 October 2026 — methodology-only analytical continuation

Status. T.A is approved. The new Q.J.F.P question concerns the time meaning of competition proportions, not a held source rate. Both alternatives have qualified local examples; no whole joined solver or gradient is adopted.

1. T.A state, units and conservation

Use metres, seconds and continuous vehicle-equivalent mass (eq). Density ρ has units eq/m and flow has units eq/s. The common triangular field is

$$F(\rho) = \min\{u\rho, w(\kappa - \rho)\}, \quad u = 30,\ w = 5,\ \kappa = 3/20, \quad \rho_c = 3/140,\ Q = 9/14. \tag{1}$$

Sections have $L = 30$ m and public interval $T = 1$ s. Sampled source sending and movement masses are

$$S_a = \min\{n_a(0), QT\}, \qquad d_{ab} = \theta_{ab} S_a, \quad \sum_b \theta_{ab} = 1. \tag{2}$$

The choice fractions use the approved prior-period lookahead Edie observations; they are not newly sampled at interior wave events. For actual accepted port flow f_{ab}, define the unspent numerical claim

$$b_{ab}(t) = d_{ab} - \int_0^t f_{ab}(s)\, ds \geq 0. \tag{3}$$

This is request accounting, not a second traffic population or assigned vehicle identity. Pair each transfer once. When a claim reaches zero, its movement cannot discharge further this interval. There is no imposed bound $f_{ab} \leq d_{ab}/T$. For constant local rate $f > 0$, the remaining amount is used after b/f seconds; stop at that exact event rather than an arbitrary threshold. Claims refresh at the public clock, without rerouting unspent mass.

The donor's total outflow uses its own physical field. All target inputs spend one receiving resource, and outgoing debits must not spend donor mass twice. Under the existing upstream-port geometry and $L = uT$, post-clock incoming traffic cannot traverse a whole section within the interval. Consequently

$$\sum_b \int_0^T f_{ab}\, dt \leq \min\{n_a(0), QT\} = S_a. \tag{4}$$

Individual d_{ab} do not follow from that total bound. They need the claim ledger. E1 accumulates completed actual transfers and creates zero-population effects at the public clock, never at interior claim exhaustion.

2. Exact allowance exhaustion and stop waves

One donor starts with jam density on $[86/3, 30]$ and vacuum upstream; its mass is $1/5$. Upstream admission is closed. Desired lateral and through masses are $1/20$ and $3/20$. Use sampled lateral choice fraction $1/4$; the through target has zero receiving, while the lateral target is free. Before the claim is fulfilled, the source-oriented local rate is

$$\beta = Q/4 = 9/56, \quad \rho_\beta = \kappa - \beta/w = 33/280, \quad c_\beta = \beta/\rho_\beta = 15/11. \tag{5}$$

It is a declared source-orientation regime, not a unique general allocator selected by T.A. The outgoing queue trace has sending Q. The initial jam-to-ρ_β release wave travels upstream at -5, reaching the rear at

$$t_0 = (4/3)/5 = 4/15. \tag{6}$$

Before t_0, vacuum ends at $86/3$, then jam extends to $30 - 5t$, followed by ρ_β to the exit. Afterwards the vacuum-to-ρ_β rear travels at c_β. The lateral claim is fulfilled at

$$t_q = (1/20)/\beta = 14/45, \quad x_q = 86/3 + c_\beta(t_q - t_0) = 316/11. \tag{7}$$

At t_q, stop lateral flow. Through receiving is still zero, so donor outflow becomes zero without reassigning its remaining through claim. A jam wave travels backward from 30 at -5, meeting the moving rear after

$$t_c - t_q = \frac{30 - x_q}{5 + c_\beta} = 1/5, \qquad t_c = 23/45, \quad x_c = 29. \tag{8}$$

Thereafter the vacuum-to-jam rear is stationary. At $T = 1$ the donor is jammed on $[29, 30]$, with mass $3/20$. Exactly $1/20$ transferred. At service start and stop use the post-event one-sided port trace: its new wave initially has zero length, so this changes no integrated field mass. In the initially empty recipient, density is $\beta/u = 3/560$ on $[u(1 - t_q), u] = [62/3, 30]$ at the completed clock. Its mass is $1/20$; no positive-measure traffic has exited it before that clock. Donor/recipient counts therefore pair exactly.

Without the T.A amount check, repeated source-oriented rate β could continue for the entire second: remaining donor mass $1/5 - \beta = 11/280 > 0$. The ordinary rear then ends at $89/3$, with ρ_β on its last $1/3$ m. Accepted lateral mass would be $9/56 > 1/20$, although aggregate traffic is still conserved. This refutes a naive repeated-demand transplantation, not native finite IT or continuous mass. This example resolves concrete wave interactions at depletion; arbitrary depleted multi-port modes remain unqualified.

3. Actual ramp/two-lane geometry for the allocation comparison

Donor 1 advances through to shoulder recipient A. Donor 2 advances laterally to A with sampled fraction $\theta = 1/4$, or through to free recipient B with fraction 3/4. A finite KW ramp also enters A. There is no same-section lateral transfer followed by another longitudinal advance.

Initially donor 1 is uniformly free at $\rho_1 = 1/150$, with sending $D_1 = 1/5$; donor 2 and the ramp start critical, sending $D_2 = D_r = Q$. Compatible prescribed inlet forcing is $1/5, Q, Q$, respectively. These are physical boundary inputs; the ramp admission can be realized by M.A under adequate offered flow and permission. No downstream wave reaches its upstream meter in this one-second case. Use supplied prior Edie observations such that $\theta = 1/4$ (e.g. 30 and 15 m/s with relaxation time 2 s). Their reached control history is not claimed.

Recipient A has density $\rho_A = 9/100$ on $[0, 1]$ and vacuum ahead. An old zero-passing effect starts at 1, with held desired speed $s = 10/3$. Own ahead traffic is vacuum, so K.A permits exactly s. The queue grows with the record:

$$z(t) = 1 + \tfrac{10}{3}t, \quad F(\rho_A) = s\rho_A = 3/10. \tag{9}$$

Its own inlet supply $R = 3/10$ is constant. This receiving history is derived from the same queue/record field, not an external capacity penalty. With illustrative fixed ramp share $p = 1/3$, both groups request more than their shares. N.A awards

$$f_r = 1/10, \qquad R_m = 1/5 \quad \text{for all non-ramp inputs together.} \tag{10}$$

Recipient B starts free at density $(27/56)/30 = 9/560$, with free exit; its inflow/outflow is 27/56. All allowances are as sampled:

$$d_{1A} = 1/5, \quad d_{2A} = 9/56, \quad d_{2B} = 27/56, \quad d_{rA} = 9/14. \tag{11}$$

No movement is fulfilled before the interior times considered below; d_{2B} is exactly fulfilled at the ending clock. Thus the allocation distinction exists even before a quota-depletion policy is needed.

P.A. Retain sampled non-ramp proportions $1/5 : 9/56 = 56 : 45$ as competition weights. Its accepted rates are

$$f^A_{1A} = 56/505, \quad f^A_{2A} = 9/101, \quad f^A_{2B} = 27/56. \tag{12}$$

P.B. Solve the source-oriented wave junction consistently. Both donors are restricted and therefore have congested boundary sending Q. Its oriented proportions are $Q : (Q/4) = 4 : 1$, giving

$$f^B_{1A} = 4/25, \quad f^B_{2A} = 1/25, \quad f^B_{2B} = 27/56. \tag{13}$$

Both give the same R_m and respect all amount ceilings.

4. Compatible own fields, counts and births for both allocations

For candidate $k \in \{A, B\}$, donor 1's queue trace is $\rho^*_{1k} = \kappa - f^k_{1A}/w$. Its shock from the initial free state has speed

$$\sigma_{1k} = \frac{f^k_{1A} - 1/5}{\rho^*_{1k} - 1/150} < 0. \tag{14}$$

It lies between $-w$ and u. At $t = 1$ donor 1 has its initial density on $[0, 30 + \sigma_{1k}]$ and ρ^*_{1k} to its port. The wave never reaches the inlet. Its own sending is then Q; this is distinct from actual service. Donor 2's total outflow is $g_{2k} = f^k_{2A} + 27/56 < Q$, with queue density $\rho^*_{2k} = \kappa - g_{2k}/w > \rho_c$. Its shock travels at -5, leaving initial critical density on $[0, 25]$ and ρ^*_{2k} on $[25, 30]$. The ramp similarly has queue density $\kappa - (1/10)/w = 13/100$ on $[25, 30]$.

Direct field integration equals paired conservation in every lane:

Ending mass	P.A	P.B
Donor 1	146/505	6/25
Donor 2	4041/5656	1069/1400
Ramp	83/70	83/70
Recipient A	39/100	39/100
Recipient B	27/56	27/56

The two donor-mainline masses sum to 281/280 under both. Initial mass plus inlets minus external exits equals the total ending inventory; internal transfers cancel. Descriptors add no population.

For recipient A, local count gauge is

$$C_A(t) = 9/100 + (3/10)t, \quad N_A(t, x) = C_A(t) - \int_0^x \rho_A(t, y)dy. \tag{15}$$

The old record has $N_A(t, z(t)) = 0$ throughout. At $t = 1$, $z = 13/3$, the queue mass and gauge both equal 39/100. Thus both own-field motion/count histories are consistent, although entry composition differs.

For an illustrative eligible lateral counter $H_0 = 47/50$ and threshold one,

$$H_A(1) = 47/50 + 9/101 = 5197/5050 > 1, \qquad H_B(1) = 47/50 + 1/25 = 49/50 < 1. \tag{16}$$

P.A creates one new E1 effect, attached at recipient A's completed inlet count 39/100; P.B creates none. The distinction is not a same-law hidden-information witness: these are different temporal allocation completions.

5. A local P.B wave-junction completion beyond naive substitution

Derive a one-fork/two-recipient kernel with $0 < \theta < 1$, two positive input demands $D_i \le Q_i$, recipient B unconstrained, receiving R_m at A, and pending amount claims. This is a reviewer derivation from source IT and ordinary boundary-wave admissibility, not a generic FIFO theorem. Here "B unconstrained" means its supply is at least $(1-\theta)Q_2$; this holds for the common-capacity free recipient in the example.

Let $\gamma \in [0,1]$ be A's common IT receiving factor and ξ_i auxiliary interior demands. Input 1 sends $\gamma\xi_1$. Input 2 sends $\theta\gamma\xi_2$ to A and $(1-\theta)\xi_2$ to B. A restricted incoming road's stationary boundary state is congested: its interior sending is capacity. An unrestricted incoming road retains total accepted flow equal to D_i; an admissible interior state can have demand between D_i and Q_i. Therefore

$$f_{1A}(\gamma) = \min\{D_1, \gamma Q_1\}, \tag{17}$$

$$\xi_2(\gamma) = \min\left\{Q_2, \frac{D_2}{1-\theta+\theta\gamma}\right\}, \tag{18}$$

$$f_{2A}(\gamma) = \theta\gamma\xi_2(\gamma), \qquad f_{2B}(\gamma) = (1-\theta)\xi_2(\gamma). \tag{19}$$

Define $H(\gamma) = f_{1A}(\gamma) + f_{2A}(\gamma)$. If $R_m \ge H(1) = D_1 + \theta D_2$, use $\gamma = 1$. Otherwise $H(0) = 0$ and H is continuous and strictly increasing, so solve $H(\gamma) = R_m$ uniquely. The second term has positive derivative: θQ_2 on its capacity branch, or $\theta D_2(1-\theta)/(1-\theta+\theta\gamma)^2$ on its demand branch. The first term is nondecreasing. All input totals are at most their incoming demands, and all output flows obey receiving.

At a restricted input, total accepted flow is below D_i, so its capacity branch is already selected. Replacing that input's demand by capacity after its queue forms leaves the same selected flow. This is the local consistency that a direct initial point-proportional substitution lacked.

For the actual comparison $D_1 = 1/5$, $D_2 = Q = 9/14$, $\theta = 1/4$, $R_m = 1/5$, both capacity branches give

$$\gamma = \frac{R_m}{Q(1+\theta)} = 56/225, \quad f_{1A} = 4/25 < 1/5, \quad f_{2A} = 1/25, \quad f_{2B} = 27/56. \tag{20}$$

Input 2's total $731/1400$ is below Q, consistent with its queue. The source rule holds on the congested interior states. Input 1's actual traffic field receives the admissible shock in section 4. No externally prescribed queue or altered ramp priority is needed to select these local flows.

Limits. Two constrained recipients, more lanes/inputs, zero or unit choice fractions, depleted claims and arbitrary restriction/node contacts require separate completion. This algebra is not an all-network theorem or a convergence proof for the finite source step. Primary continuum general-junction theorems cited here include FIFO coupling; they are not imported.

6. P.A's local fixed-proportion completion and the pending choice

For the same one-fork geometry, let sampled weights $\mu_1, \mu_2 > 0$ remain constant within the public interval and $\alpha = \mu_1/(\mu_1 + \mu_2)$. The same unconstrained-B supply condition applies. An explicitly declared local fixed-priority completion uses potential target-A demands D_1 and θD_2. With unused shares transferred, define

$$f_{1A} = \min\{D_1, \max(\alpha R_m, R_m - \theta D_2)\}, \tag{21}$$

$$f_{2A} = \min\{\theta D_2, \max((1-\alpha) R_m, R_m - D_1)\}, \tag{22}$$

$$\xi_2 = \min\left\{Q_2, \frac{D_2 - f_{2A}}{1-\theta}\right\}, \quad f_{2B} = (1-\theta)\xi_2. \tag{23}$$

Since $f_{2A} \le \theta D_2$, $\xi_2 \ge D_2$, and the donor total $f_{2A} + f_{2B}$ is at most D_2. If that total is strictly below D_2, then $\xi_2 = Q_2$ and the donor queue is admissible. If equal, an interior demand $\xi_2 \in [D_2, Q_2]$ is admissible. Thus this kernel also has compatible local wave states, without introducing a held d/T rate ceiling.

In the comparison, $\mu_1 = 1/5$, $\mu_2 = 9/56$, $\alpha = 56/101$; both shares are below their potential demands, giving $56/505$ and $9/101$. The second donor is capacity-restricted in total, so $\xi_2 = Q$ and its parallel through flow is $27/56$, as in the exact field table.

In this local geometry, the fixed-weight kernel is invariant to a restricted input's demand increase. For input 1 this is the usual fixed-priority merge property. For input 2, strict total restriction implies $f_{2A} < D_2 - (1-\theta)Q_2 \le \theta D_2$; its target-A demand cap is already nonbinding in the allocation. Increasing D_2 to Q_2 leaves its assigned share unchanged, while the same parallel capacity branch is retained. Unrestricted inputs do not require a queue-induced demand replacement.

This is a declared source-time priority adaptation, not a proof that native IT already specifies it for nonuniform physical-time fields. The primary node literature supports fixed-priority allocation; it does not choose our sampled-demand weights or qualify every joined/depleted mode.

Choice. T.A determines each movement's integrated ceiling. It does not choose whether its original request determines a standing service weight (P.A), or source-proportional wave-junction demands determine its share (P.B). Both local kernels can be causal, conservative and quota-compatible, yet the flows and next-clock effects differ. Q.J.F.P therefore reopens only the temporal meaning of IT competition, including N.A's non-ramp subgroup. Fixed ramp priority/shared receiving, M.A/G1/E1/K.A/T.A stay approved.

P.A is provisionally preferred for the sampled task. After selection, qualify shared-donor/multi-target allocation and depleted modes, then the remaining motion/contact/crossing/E1 completion. No full Q, clearance, full TST/control response, useful gradient or Stage 4 is claimed.

Primary reading and attribution

Laval and Daganzo (2006), equations 7–13: sampled oriented amounts and IT. Wright et al., arXiv:1601.01054, sections 2.1/3.1: demand, priority and invariance; arXiv:1608.07623: cumulative node service/completion events, whose snapshot execution clock is not automatically our physical clock. Jin, arXiv:1204.6727, section 3.1: admissible stationary/interior states; its general theorem has a different FIFO scope. Sections 2–6 above contain reviewer derivations under their stated conditions.

Source IT Priority/node manuscript Node dynamic-system manuscript Wave-junction manuscript